DRUGS · ANTIBIOTICS: OTHER CLASSES

Clarithromycin

Clarithromycin is the macrolide an ENT surgeon uses most: the usual alternative for penicillin-allergic patients with otitis media, sinusitis or tonsillitis, and, given at low dose for three months, an anti-inflammatory treatment for chronic rhinosinusitis without polyps. It is a potent inhibitor of the liver enzyme CYP3A4, so its interactions matter more than its spectrum.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

CLARITHROMYCIN

	SUMMARY
Class	Macrolide
How it works	Binds the 50S ribosomal subunit and blocks protein synthesis; bacteriostatic; also reduces neutrophil activity, mucus hypersecretion and biofilm formation
Spectrum	Streptococci (resistance in a minority of pneumococci and group A streptococci), Moraxella, H. influenzae (moderate), atypicals (Mycoplasma, Chlamydophila, Legionella), Bordetella, non-tuberculous mycobacteria, Helicobacter
ENT uses	Penicillin allergy: acute otitis media, sinusitis, tonsillitis; long-term macrolide therapy for chronic rhinosinusitis without polyps; non-tuberculous mycobacterial lymphadenitis (with rifampicin or rifabutin) when excision is not possible; pertussis
Adult dose	250 to 500 mg twice daily for 5 to 7 days; long-term CRS regimen 250 mg once daily (or twice daily) for 12 weeks
Child dose	7.5 mg/kg twice daily (maximum 500 mg), or by weight band: 8 to 11 kg 62.5 mg, 12 to 19 kg 125 mg, 20 to 29 kg 187.5 mg, 30 to 40 kg 250 mg, twice daily
Avoid	QT prolongation, hypokalaemia; with simvastatin, lomitapide, colchicine in renal impairment, ergotamine, ticagrelor

	SUMMARY
Pregnancy	Avoid, especially in the first trimester, unless benefit clearly outweighs risk; erythromycin is preferred

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Antibacterial and anti-inflammatory action

At usual doses it stops bacterial protein synthesis at the 50S subunit. At low doses given for months, macrolides reduce interleukin-8 and neutrophil recruitment, reduce goblet-cell mucus secretion and interfere with quorum sensing and biofilms; this immunomodulatory effect, not killing, is the rationale in chronic rhinosinusitis.

2.2 Handling

Well absorbed, concentrates in tissues and macrophages; metabolised by CYP3A4 to an active metabolite and excreted partly in urine. Halve the dose when the eGFR is below 30 ml/min.

3 CHAPTER THREE Using it in ENT

Indications and doses.

3.1 Acute infection in penicillin allergy

NICE lists clarithromycin (or erythromycin, and doxycycline in adults) as the alternative in penicillin allergy for acute otitis media, acute sinusitis and sore throat.

3.2 Chronic rhinosinusitis

Long-term low-dose macrolides are considered for **chronic rhinosinusitis without polyps** with low IgE and neutrophilic inflammation, after maximal medical therapy, as in the EPOS and BRS guidance: 250 mg once daily for 12 weeks, continued only if it helps. Benefit is modest and less clear in eosinophilic polyposis. Check an **ECG for QT** and a statin and drug history before starting.

3.3 Mycobacterial lymphadenitis

Non-tuberculous mycobacterial node infection in young children is best treated by **complete excision**; where excision would risk the facial nerve, clarithromycin with rifampicin or rifabutin for 3 to 6 months is an alternative.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- Nausea, diarrhoea, abdominal pain and a metallic taste (common).
- **QT prolongation** and torsades de pointes, especially with other QT-prolonging drugs or hypokalaemia.
- Reversible hearing loss and tinnitus at high doses.
- Hepatotoxicity; C. difficile; increased cardiovascular mortality was reported years after a course in patients with coronary heart disease, so use with caution in them.

4.2 Interactions

A strong **CYP3A4 inhibitor**: contraindicated with simvastatin (rhabdomyolysis), and raises levels of atorvastatin, ciclosporin, tacrolimus, carbamazepine, theophylline, midazolam, DOACs, calcium-channel blockers and warfarin (raised INR). Additive QT effects with antipsychotics, quinolones, ondansetron and antiarrhythmics.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The **anti-inflammatory macrolide** rationale and who benefits in chronic rhinosinusitis.
- The key **interactions** (statins, warfarin, QT drugs) and the ECG before long-term use.
- Management of **atypical mycobacterial lymphadenitis** in a child.

5.2 Common pitfalls

COMMON PITFALLS

- Starting long-term clarithromycin in a patient on simvastatin.
- No ECG before long-term therapy in a patient with cardiac disease or on QT drugs.
- Expecting long-term macrolides to shrink eosinophilic nasal polyps.

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